

Linear Regression Evaluation Methods

Our main goal In linear regression is identify the coeff

in order to minise the cost function

So cost function is One of our Evaluation metric

In Regression problem or Forecasting problem

We have some metrics will be avilable

Metrics will provide the perfomance of the model , that we are calling as

Model evaluation

- MSE or Cost (Mean Square Error)
- RMSE (Root Mean square Error)
- RSE (Residual Standard Error)
- R – square
- Adjusted R square
- AIC: Akaiki Information creteria
- BIC: Baysian Information Creteria

MSE: Mean square error

$$MSE = \frac{1}{n} * SSE$$

$$MSE = \frac{1}{n} * \sum_{i=1}^n (y_{a_i} - y_{p_i})^2$$

ROOT MEAN SQUARE ERRORS (RMSE)

$$RMSE = \sqrt{MSE}$$

$$RMSE = \sqrt{\frac{1}{n} * \sum_{i=1}^n (y_{a_i} - y_{p_i})^2}$$

RSE: Residual Standard error

Statistics part1 – Standard deviation

How much an observation deviated from population mean

Statistics part2 – Standard error

How much a sample mean deviated from population mean

Linear regression: Residual standard error

How a sample regression line is deviating from population regression line

Here also we divided our data into samples

each sample we will get one regression line

that regression never equal to population regression line

it has some deviation, that deviation we can measure in terms of RSE

In other words RSE will give some error even though when we have a best fit line

RSE is irreducible error

RSE will calculate after we got best fit line

$$\text{Standard deviation} = \sigma = \sqrt{\text{var}} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\text{Standard Error} = \frac{\sigma}{\sqrt{n}}$$

$$RSE = \sqrt{\frac{1}{n - p - 1} \sum_{i=1}^n (y_{\text{actual}} - y_{\text{predicted}})^2}$$

$$\sum_{i=1}^n (y_{\text{actual}} - y_{\text{predicted}})^2 = SSE = RSS$$

$$RSE = \sqrt{\frac{RSS}{n - p - 1}}$$

p = predictors = number of input columns

n = number of observation

$$RSE = \sqrt{\frac{1}{n - p - 1} \sum_{i=1}^n (y_{actual} - y_{predicted})^2} = \sqrt{\frac{RSS}{n - p - 1}}$$

$$Sales = b_0 + b_1 * TV + b_2 * SM + b_3 * NP$$

$$RSE = \sqrt{\frac{RSS}{n - 3 - 1}} = \sqrt{\frac{RSS}{n - 4}}$$

RSE is also called as lack of fit

it gives the deviation between fitted line to population lines

RSE will not give indetaile(which column more effect which column less effect)

between how an independent variable

affecting the predictions

R - square

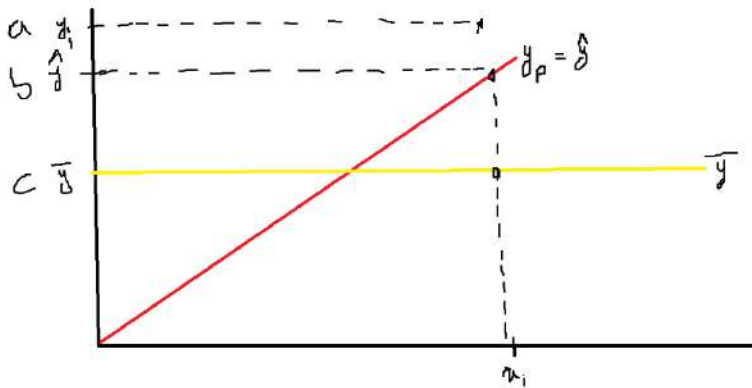
$\hat{y}_i$ or y_p = model predictions, y_i = actual output, $\bar{y}_i$ = average actual output value

$$SS_{Total} = SS_{Residuals} + SS_{Explained}$$

SS_{Total} = Total Sum of squares

$SS_{Residuals}$ = Residual sum of squares

$SS_{Explained}$ = Explained sum of squares (un explained error)



y_{ai} = actual value

$\hat{y}_i$ = prediction due to regression line

$\bar{y}$ = avg value of all actual output

$$\sum_{i=1}^n (y_{ai} - \bar{y})^2 = \sum_{i=1}^n (y_{ai} - \hat{y}_i)^2 + \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

$$SST = \sum_{i=1}^n (y_{ai} - \bar{y})^2$$

$$SSR = \sum_{i=1}^n (y_{ai} - \hat{y}_i)^2$$

$$SSE = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

$\hat{y}_i - \bar{y} = \text{explained error after LR and average value}$

$$SS_{Total} = SS_{Residuals} + SS_{Explained}$$

$$SS_{Total}/SS_{Total} = SS_{Residuals}/SS_{Total} + SS_{Explained}/SS_{Total}$$

$$1 = \frac{SS_{Residuals}}{SS_{Total}} + \frac{SS_{Explained}}{SS_{Total}}$$

$$\frac{SS_{Explained}}{SS_{Total}} = 1 - \frac{SS_{Residuals}}{SS_{Total}}$$

$$R - \text{squared} = 1 - \frac{SS_{Residuals}}{SS_{Total}}$$

$$R - \text{squared} = 1 - \frac{\text{sum of squared residuals (SSR)}}{\text{Total sum of squares (SST)}}$$

$$R - \text{squared} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

$$R - \text{squared} = 1 - \frac{\text{Error due to Regression model}}{\text{Error due to Mean model}}$$

R – squared Interpretation

- *R – squared is called as **Coefficient of Determination***
- *R – squared is also called as **Good ness of fit, RSE: Lack of fit***
- *R_{sq} explained how much variability in output column based on input column*
- *How much percentage of variance in output column explained by input column*

- $R - squared$ is varies from 0 to 1
- More R_{Square} value indicates good model
- R_{Square} gives the idea of *explainability of model*

For example

$$sales = b_0 + b_1 * TV$$

assume that by using OLS method we identify the coeff values

$$sales = 1000 + 20 * TV \text{ =====> LR model}$$

Evaluated the model

$$\text{Model is Giving } R_{Square} = 0.8$$

80% of variance in sales is explained by TV

$$son = b_0 + b_1 * mama$$

90% of variance in son is explained by mamma

$$son = b_0 + b_1 * mama + b_2 * papa$$

95% of variance in son is explained by mamma

$$son = b_0 + b_1 * mama + b_2 * papa + b_3 * friend$$

100% of variance in son is explained by mamma

Note:

R_{Square} value keep on increase when you add the variable

$$R - squared = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

X	Y = y_i	$\hat{y}_i = 0.143 + 1.229x$	$y_i - \hat{y}_i$	$(y_i - \hat{y}_i)^2$	$y_i - \bar{y}$	$(y_i - \bar{y})^2$
2	2					
3	4					
4	6					
6	7					
	$\bar{y}$ =			$\sum_{i=1}^n (y_i - \hat{y}_i)^2$		$\sum_{i=1}^n (y_i - \bar{y})^2$

X	Y = y_i	$\hat{y}_i = 0.143 + 1.229x$	Residual $y_i - \hat{y}_i$	Square of residual $(y_i - \hat{y}_i)^2$	$\bar{y}$	$y_i - \bar{y}$	$(y_i - \bar{y})^2$
2	2	2.601	2-2.601=- 0.601	0.361201	4.75	2-4.75=- 2.75	7.5645
3	4	3.83	4-3.83=0.17	0.0289	4.75	4-4.75=- 0.75	0.5625
4	6	5.059	6- 5.059=0.941	0.885481	4.75	6- 4.75=1.25	1.5625
6	7	7.517	7-7.517=- 0.517	0.2672	4.75	7- 4.75=2.25	5.0625
			SSR=	1.542871			14.75

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2 = 1.542871$$

$$\sum_{i=1}^n (y_i - \bar{y})^2 = 14.75$$

$$R - squared = 1 - \frac{1.542871}{14.75} = 0.8954$$

number lectures per day affects the number of hours spent at university per day

$no_of_lectures = 0.143 + 1.229 * \text{Hours spent}$

$$R_{squared} = 0.8954$$

89% of variance in lecture explain by hours

The number of lectures are varying w.r.t hours , is that True ?

Yes

Problem using R_{square}

we already know R_{square} value keep on increase when we add variable in the model

For example a data has 100 columns

You used 50 columns :

$$y = b_0 + b_1 * x_1 + b_2 * x_2 + \dots + b_{50} * x_{50} \implies R_{square} = 0.8$$

You used 60 columns :

$$y = b_0 + b_1 * x_1 + b_2 * x_2 + \dots + b_{60} * x_{60} \implies R_{square} = 0.9$$

You used 100 columns :

$$y = b_0 + b_1 * x_1 + b_2 * x_2 + \dots + b_{100} * x_{100} \implies R_{square} = 1$$

whenever we have 100% accuracy , then the model is called as overfit